澳門大學
UNIVERSIDADE DE MACAU
UNIVERSITY OF MACAU

DiffOSeg: Consensus and Preference

Keru ZHENG Yanchen MU Huanrong LIU

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1.1 Multi-rater segmentation



One CT patch · four expert annotations

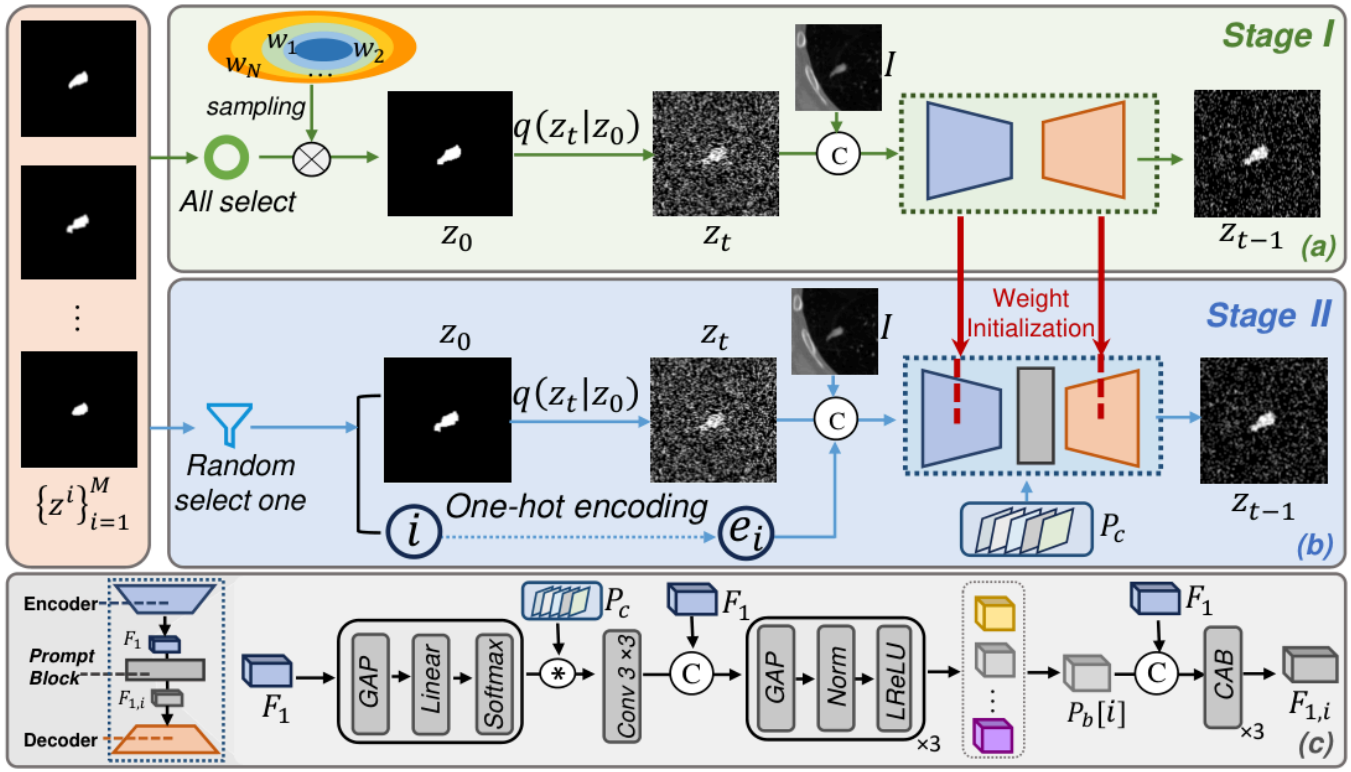
Consensus

Which segmentation patterns are shared, and where do annotations disagree?

Preference

How can a prediction reflect a particular annotation style?

1.2 DiffOSeg in two stages



Stage I

Image-conditioned categorical diffusion learns a distribution of masks.

Stage II

Initialize from Stage I; add adaptive preference prompts.

Our target: the Stage I consensus strategy.

1.3 P versus A

P: probabilistic consensus

Choose 1–4 annotations, then average the selected subset.

$$z_P = \frac{1}{|S|} \sum_{j \in S} z_j$$

The training target changes with the subset.

A: average aggregation

Average all four annotations on every visit.

$$z_A = \frac{1}{4} \sum_{j=1}^{4} z_j$$

The consensus probabilities stay fixed.

Frozen local choices: uniform subset size and subset; per-pixel categorical target draws in both arms. These operational details are disclosed assumptions.

Does P outperform A on LIDC-IDRI?

Table 3 · Stage I · scaled-faithful · one patient fold · three paired seeds

Endpoint	What it measures	Expected change
GED30	Mismatch between mask distributions	$P - A < 0$
DiceSoft30	Threshold-aware overlap of mean masks	$P - A > 0$

30 = generated masks per image. DiceSoft uses five thresholds.

Supported: both paired 95% intervals favor P.
Not supported: either metric is significantly reversed.
Inconclusive: all other outcomes.

2.2 Paired experiment

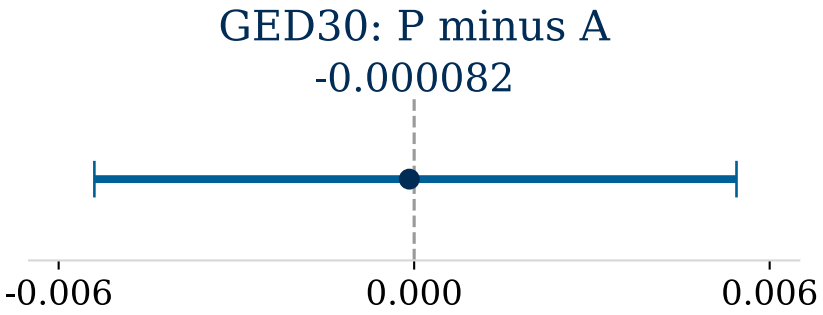
Design choice	Frozen local comparison
Patient separation	160 train / 54 test patients; 450 test patches
Repeated training	P and A × 3 paired seeds; matched initialization
Shared budget	120,000 updates; batch 12; final EMA
Preserved settings	128 × 128; 250 cosine steps; K = 30
Compute	12.157 GPU-hours used / 240-hour ceiling

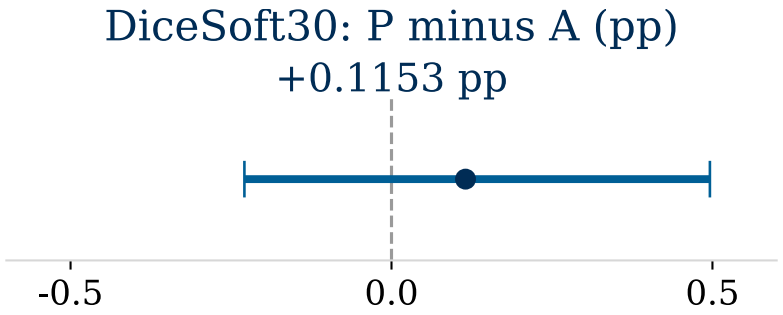
Paper: four folds. Local test: one fold, with frozen consensus encoding and augmentation assumptions.

Earlier P-only runs had already examined this cohort. The freeze predates this P/A comparison, not all prior exposure.

2.3 P/A result: inconclusive

Metric	Paper P / A	Local P / A
GED30 ↓	0.0773 / 0.1029	0.104276 / 0.104358
DiceSoft30 ↑	92.20 / 90.43%	90.3139 / 90.1986%





Both paired intervals cross zero: the declared outcome is inconclusive.

95% intervals use 10,000 paired seed-by-patient draws; patients retain all patches. Paper: four-fold mean. Local: three-seed mean on one fold.

2.4 Discrepancy diagnosis

Checks completed

- Matching split, budget and paired initialization
- Source and checkpoint identities checked
- Saved mask metrics and bootstrap reproduced

Explanations still open

- Consensus encoding and augmentation choices
- Split composition and training variability
- One-fold scale and prior cohort exposure

Under our sampler, $E[z_P] = z_A$ at each pixel; shared subsets can still change spatial dependence.

This mechanism does not identify the cause of the paper–local gap. An inconclusive result does not prove equivalence.

3.1 Hypotheses and pilot gates

Separate four-fold extensions of Reproduced DiffOSeg; pilots use outer-training patients.

Hypothesis / limitation	Quantitative inner evidence	Rule / next step
H1 · Boundary Localization error	Dice 89.06–89.80%	≥80% in every fold ⟶ v1
H2 · Polar Contour variation	Dice 86.33–87.18%	≥78% in every fold ⟶ v2
H3 · Votes Pixel disagreement	ΔGED50 −0.004412 ΔDiceSoft50 +1.611 pp	12 options/fold; maximize minimum relative gain ⟶ v3
H4 · Energy Distribution mismatch	Decoder-only fails; energy readout passes	Three-metric gate ⟶ v4 Details on the next slide

H1/H2: standalone Dice ranges across inner folds; finite-output checks also required. H3: mean changes vs its inner surrogate; parameter selection, not a fixed cutoff.

Passing a pilot permits evaluation. v2 later failed all three outer improvement directions.

3.2 Why v4 was promoted

Gate: all three four-fold mean changes must improve

Candidate – locked v3 surrogate, K=50. Required: ΔGED < 0; both ΔDice > 0. Dice changes are percentage points.

Inner pilot	ΔGED50	ΔDiceSoft50	ΔPref. Dice	Decision
Decoder only	+0.003270	+0.131	+0.151	FAIL; no outer test
+ Energy readout	−0.006944	+0.194	+0.151	PASS ⟶ full v4

Revision: keep the 16-state rank/vote decoder; fit a frequency readout with **GED + 3 × DiceSoft loss** to calibrate the sampled masks.

Pass inner gate ⟶ lock parameters ⟶ seal outer predictions ⟶ score

This is a point-estimate gate, not a significance test. Inner checks were reused during development. We focus on v4 because the failed pilot led to a targeted revision and full controls.

3.3 Results and controls

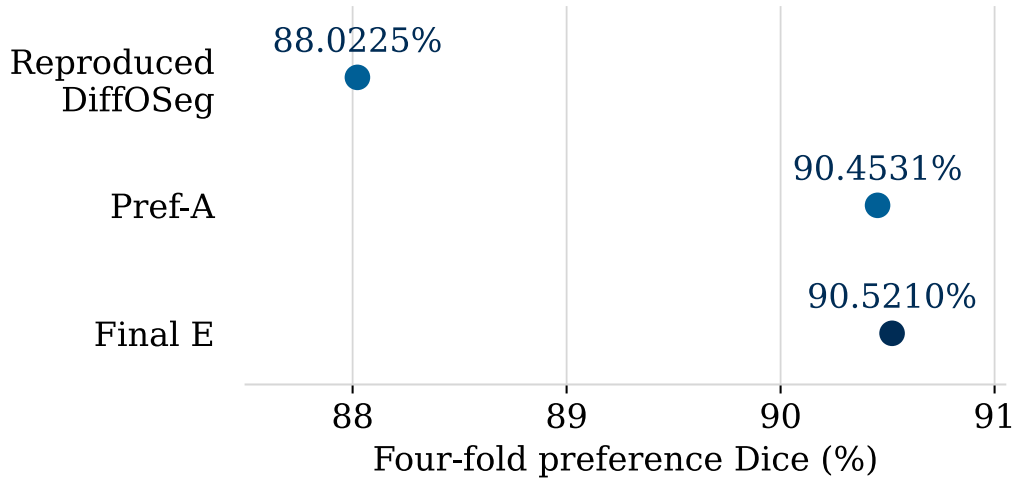
Variant	GED50 ↓	DiceSoft50 (%) ↑
Votes v3	0.107941	90.696
Full v4	0.104515	90.790
No energy readout	0.115012	90.730
Original sample order	0.103678	90.790
Energy-only on v3	0.097209	90.483

v4 – v3 GED50: –0.003426; 95% CI [–0.005314, –0.001551]

The old order improves GED further; energy-only trades DiceSoft for GED. The small v4 Dice gains have intervals crossing zero.

No-energy is a fixed-weight inference control, not the failed pilot. Patient CIs condition on fitted models; total compute is not matched.

3.4 Preference extension



90.5210%

+2.4985 pp vs Reproduced DiffOSeg

Localization / registration, followed by SAM adaptation and a three-seed SAM2.1 combination.

Extra models and compute; the historical 91% target remains unmet.

Four-fold preference Dice averages four area-ranked masks. Ranks are size-based styles, not persistent clinician identities.

Same repeatedly used cohort. Pref-A is an extension system, unrelated to A = average aggregation in the P/A test.

4.1 Agent use and evidence

Phase	Consequential action	Verification record
Understand / audit	Recovered the omitted P/A archive	Frozen source, six runs and pairing
Implement / check	Built CPU evidence checks	Metric fixtures and package checks
Analyze / challenge	Retained the inconclusive decision	Full saved-mask and bootstrap replay

Reported results are traceable; evidence checks passed.

4.2 Limits and next steps

Current limit	Next verification
P/A: one fold and only three seeds	More folds / seeds; isolate encoding assumptions
Repeated use of the LIDC cohort	Freeze the method and test a new cohort
Added models and incomplete compute totals	Account for costs; compare matched budgets
Extension CIs condition on fitted models	Repeat the complete training and selection process

A three-seed ensemble is not three independent system repetitions.

The P/A claim, four-fold consensus study and preference extension retain separate conclusions.

4.3 Conclusions

P/A reproduction: complete, but inconclusive

The local advantages are small and both paired intervals cross zero.

Energy calibration: a supported local GED gain

Negative results and component controls limit the explanation.

Preference extension: 90.5210% with added resources

Promising on this cohort; generalization and compute fairness remain open.